PAPER FOLDING

Fold Lines, Cuts and Unfolding

OAVS TGT | Reasoning (Non-Verbal) | Topic 18: ninth non-verbal topic

Syllabus line (Reasoning, 30 Marks): “The Non-Verbal Reasoning topics like Figure Formation, Dice, Triangle, Rule Detection, Images, Completion of Pattern, Mirror Images, Figure Matrix and **Paper Folding**, etc.”

Built from: OAVS TGT English 2019, English 2022, Mathematics 2024, Science 2024 and Social Studies 2023 (17 May 2023, Shift 2). Four paper-folding questions appear (English 2019 Q47, Q48, Q49 and English 2022 Q31). All four are solved with their original figures. The practice questions were generated by program so that each has exactly one option that fits. **This is the last topic named in the syllabus line.**

What is inside

Ch.	Title	What you get
1	PYQ Analysis	The 4 PYQs, the 3 question types, priority list
2	Foundation	Fold-line reflection rules, unfolding rules, hole counting, method
3	Solved PYQs	4 PYQs with original figures and step-by-step reasoning
4	Shortcuts, Traps and Scope	Speed methods and mistakes
5	Practice Set	6 original questions with a verified key
6	Quick Revision	One-page cheat sheet

CHAPTER 1

PYQ Analysis

Four questions, three types

Paper	Q.	Type	What was asked	Ans
English 2019	47	Transparent sheet, diagonal fold	Circle with an arrow pointing up-left, sheet folded on the “/” diagonal	2
English 2019	48	Fold and cut, then unfold	Paper folded twice and a strip with a star cut; select the unfolded paper	3
English 2019	49	Transparent sheet, vertical fold	Ring, star and two triangles folded on a vertical line	2
English 2022	31	Reverse: unfolded to folded	From a paper with holes, select the fold and the cuts	A
Mathematics 2024 / Science 2024 / Social Studies 2023	–		none	

PYQ INSIGHT: Three types appear: (1) **transparent sheet folded** so that the two parts overlap (2 PYQs), (2) **fold, cut, then unfold** (1 PYQ), (3) **reverse**, where the unfolded paper is given and the fold and cut must be found (1 PYQ). All use mirror reflection in the fold line. The English 2019 paper had three of the four, so expect at least one in each paper.

Question type	PYQs	Priority
Transparent sheet folded (overlay of the two parts)	2	★★★ Must
Fold and cut, then unfold	1	★★★ Must
Reverse: find the fold and cuts from the unfolded paper	1	★★★ Must
Counting holes after folds and a punch	0	★★ Expected
Diagonal fold of a punched paper	0	★ Possible

CHAPTER 2

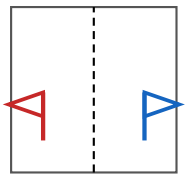
Foundation

Reflection in the fold line

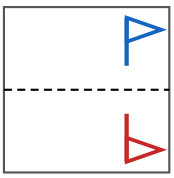
2.1 The core idea

A fold is a **mirror in the fold line**. Whatever is on the folded part is reflected across the line and lands on the other part. Folding is not turning the paper over by rotating it, so a shape changes by a reflection, never by a rotation.

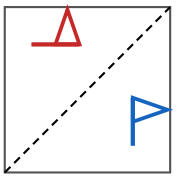
2.2 Where a figure goes (transparent sheet)



Vertical fold: left-right flip



Horizontal fold: up-down flip



Diagonal "/" fold: x and y swap

Blue is the original flag; red is where it lands after the fold. The flag keeps its shape but is reflected, so its pennant flips with it.

Fold line	Position of a point	Arrow directions change as
Vertical	Left and right swap; height unchanged	E and W swap; N and S stay
Horizontal	Top and bottom swap; left-right unchanged	N and S swap; E and W stay
Diagonal "/" (bottom-left to top-right)	x and y swap	N and E swap; S and W swap; NW and SE swap; NE and SW stay
Diagonal "\" (top-left to bottom-right)	x and y swap with signs reversed	N and W swap; S and E swap; NE and SW swap; NW and SE stay

NOTE: Symmetric figures (a circle, a ring, a six-pointed star) look the same after reflection. Only asymmetric parts (arrow, triangle with a right angle, flag) show the change, so track those first.

2.3 Fold, cut and unfold

Where the cut is	After unfolding
Inside a layer, away from every fold	The cut and its mirror image(s): one copy per layer, reflected in each fold line used
Touching a fold edge	The cut joins its mirror image into **one symmetric shape** (half-circle becomes a circle, triangle becomes a diamond, half-rectangle becomes a rectangle)
Touching two fold edges (a corner)	A single shape built from the cut and its reflections in both folds
After n folds (each doubling the layers)	An interior cut gives 2^n holes (1 fold: 2, 2 folds: 4, 3 folds: 8)

2.4 Method

Step	Do this
1	Mark the fold lines in the order they were made.
2	Unfold in **reverse order**: reflect the cut in the last fold line first, then in the one before.
3	Keep the shape's orientation reflected each time; do not rotate it.
4	Check the result: the final paper must be symmetric about every fold line used.
5	Eliminate options with the wrong count, a shape on the wrong side of a fold, or an unflipped shape.

SHORTCUT: Reverse questions: a shape lying **on** a middle line of the unfolded paper came from a cut on a fold edge; shapes lying in mirror pairs came from interior cuts. Match the position of every shape type to the fold and cut shown in the options (Eng 2022 Q31).

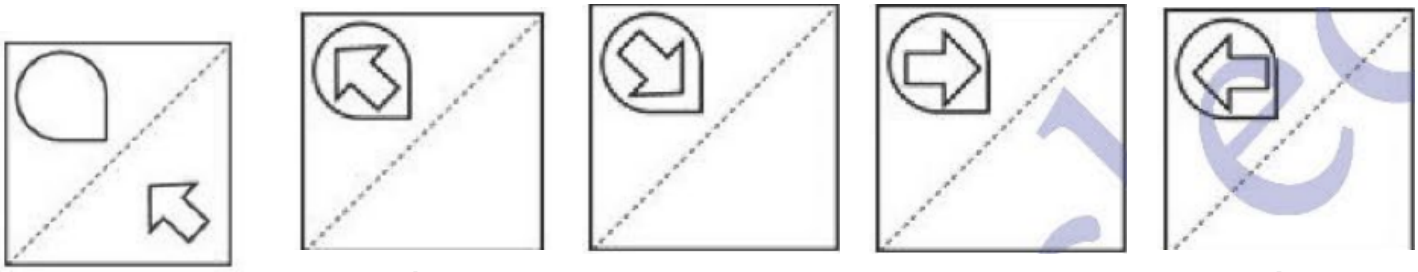
CHAPTER 3

Solved PYQs

All four questions with original figures

English 2019 Q47

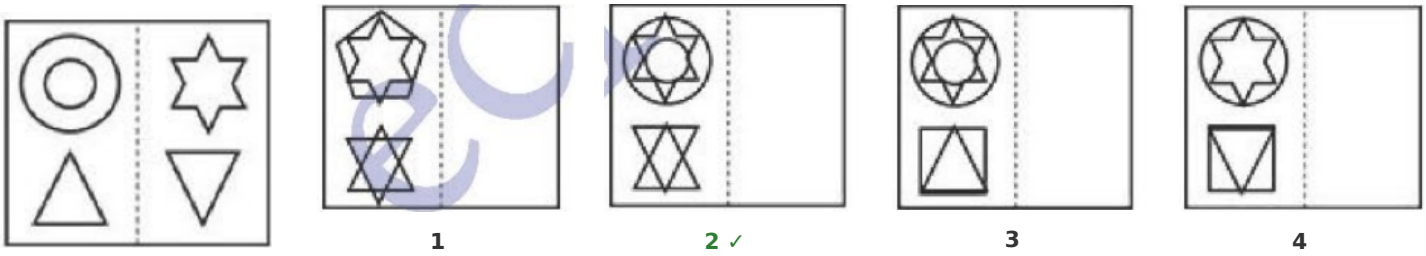
How would the pattern appear when the transparent sheet is folded at the dotted line?



1. The dotted line is the "/" diagonal. The lower-right part (the arrow) is folded over onto the upper-left part (the circle).
2. A fold is a reflection. In a "/" fold, N and E swap, S and W swap, and NW and SE swap. The arrow points up-left (NW), so it will point **down-right (SE)**.
3. Its place in the lower-right moves to the upper-left corner, inside the circle.
4. Option 1 keeps the arrow pointing up-left (not reflected). Options 3 and 4 point right and left (wrong turn). Only option 2 has a down-right arrow inside the circle.
Answer: Option 2

English 2019 Q49

How would the pattern appear when the transparent sheet is folded at the dotted line?

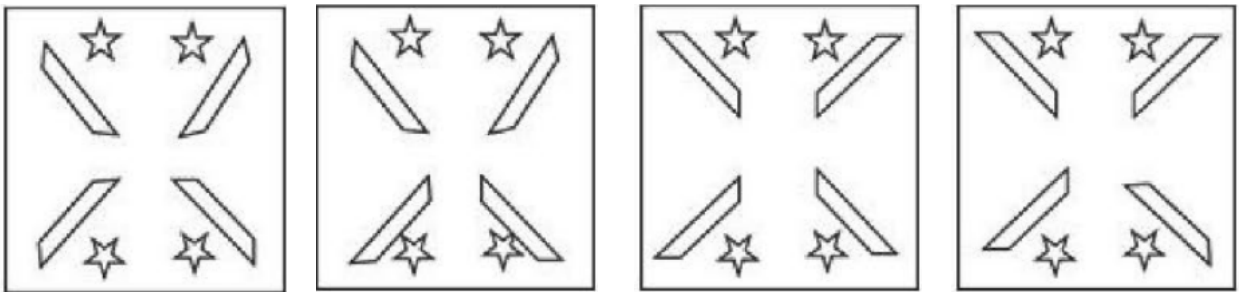


1. The dotted line is vertical, so the right half is reflected left-right onto the left half.
2. Top: the ring (circle inside circle) and the six-pointed star are both symmetric, so they overlap as a **star inside two concentric circles**.
3. Bottom: an upward triangle and a downward triangle (a left-right flip keeps up and down) overlap to make a **six-pointed star**.
4. Option 1 contains a pentagon, and options 3 and 4 contain a square, which no part of the sheet has. Only option 2 matches.
Answer: Option 2

English 2019 Q48

If the paper is folded and cut as per the shape given, what would the result look like after unfolding the paper?



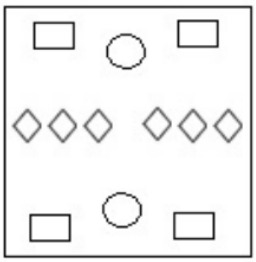


1 2 3 ✓ 4

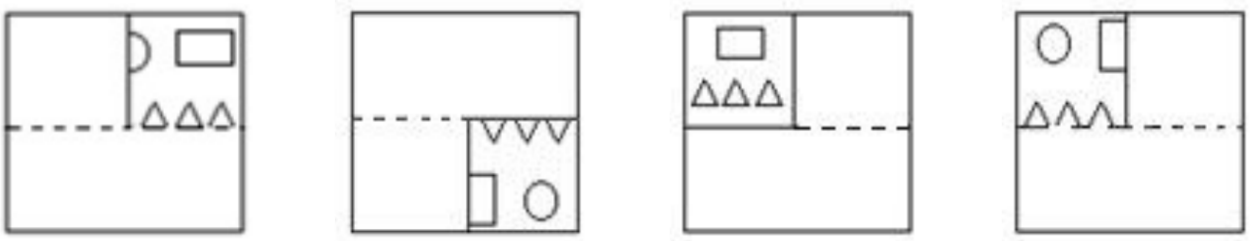
1. The paper is folded twice (the two fold lines are the middle vertical and middle horizontal lines) and a strip with a star at one end is cut in the folded quarter.
2. Unfolding reflects the cut in the horizontal fold and then in the vertical fold: the unfolded paper has **four copies**, the right pair mirroring the left pair and the bottom pair mirroring the top pair.
3. So the strips slant like "\" and "/" in the top row, "/" and "\" in the bottom row, with each star at the same end of its strip (nearer the middle line) in all four copies.
4. Option 2 has its stars overlapping the strips (the copies were not placed by mirror). Options 1 and 4 move the strips relative to the stars. Option 3 has all four copies in exact mirror positions.
Answer: Option 3

English 2022 Q31

If a paper is folded in a particular manner and punches and cuts are made, when unfolded, this paper appears as shown in the question figure. Select from among the given options the manner in which the paper is folded, and punches and cuts made.



1. The unfolded paper has: three diamonds on each side of the middle (on the horizontal middle line), a circle on the vertical middle line at the top and bottom, and a rectangle in each of the four corners.
2. Shapes **on** a middle line come from cuts on a fold edge: diamonds come from triangles on the horizontal fold edge, circles from half-circles on the vertical fold edge. Rectangles in the corners are interior cuts.
3. Option A has the half-circle on the vertical edge, the rectangle in the interior and three triangles on the horizontal edge, exactly the cuts needed.
4. Option B has the circle inside and the rectangle on the edge (the other way round); C has no half-circle on the vertical edge (no circle at the middle); D also has the circle inside and the rectangle on the edge.
Answer: Option A



A ✓ B C D

NOTE: The scan of English 2019 Q48 is small, so options 1, 2 and 4 are eliminated by the placement of strips and stars in the figure. The official key is Option 3. Q47, Q49 and English 2022 Q31 are clear from the figures.

CHAPTER 4

Shortcuts, Traps and Scope

Speed methods and mistakes

• **Reflect, never rotate:** a fold gives a mirror image of the folded part, not a turned copy.

• **Symmetry test:** after unfolding, the paper must be symmetric about every fold line; reject options that are not.

• **On-the-line shapes:** a shape on a fold line combines with its reflection into one whole symmetric shape (circle, diamond).

• **Count rule:** an interior cut through n folds gives 2^n holes.

• **Arrows:** in a "/" fold, N and E swap; in a vertical fold only E and W swap.

• **Time:** 45 seconds per question.

Trap	How to avoid it
Rotating instead of reflecting	A rotated copy looks similar but its asymmetric parts point the wrong way; test a flag-like part.
Unfolding in the wrong order	Always unfold the last fold first.
Forgetting that edge cuts merge	A half-circle on a fold edge becomes a full circle, not two half-circles.
Counting layers wrongly	Two folds give four layers, three folds give eight.
Confusing a transparent sheet with a cut paper	In a transparent-sheet question, both parts remain visible and overlap; in a cut question, only the cut shapes matter.
Missing the mirror of a diagonal fold	x and y swap in a "/" fold; N and E swap.

Scope

Related PYQs / topics	Where
Cube nets and folding a net into a cube (Eng 2019 Q55; Eng 2022 Q53)	Dice (earlier topic)
Mirror outside the figure (mirror at MN)	Mirror Images (Topic 16)

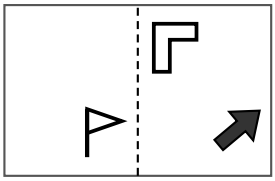
CHAPTER 5

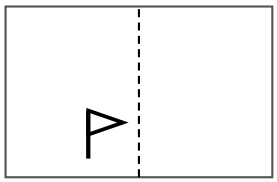
Practice Set

6 original questions; answers on the next page (every option set verified by program)

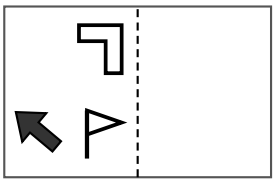
A. Transparent sheet folded at the dotted line

Q1. The transparent sheet is folded at the dotted line (right half over the left half). How will the figure appear?

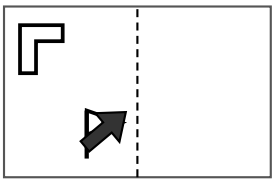




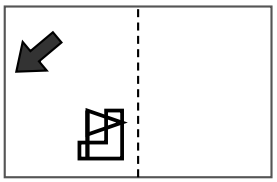
A



B

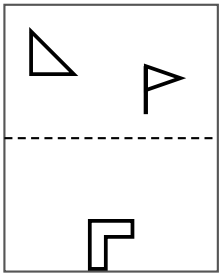


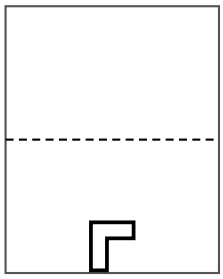
C



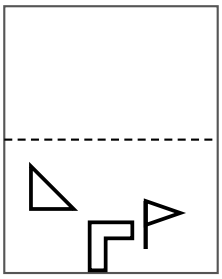
D

Q2. The transparent sheet is folded at the dotted line (top half down over the bottom half). How will the figure appear?

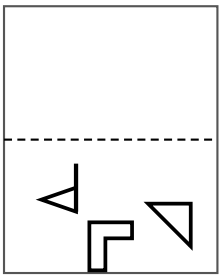




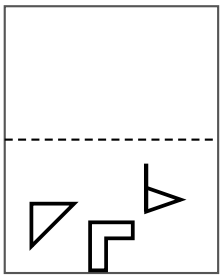
A



B

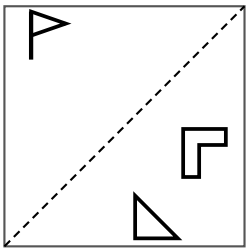


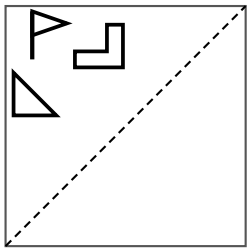
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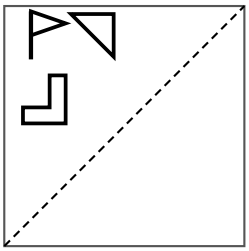
D

Q3. The transparent sheet is folded along the dotted diagonal (lower-right part over the upper-left part). How will the figure appear?

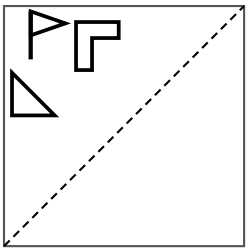




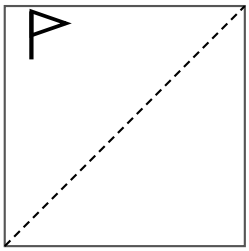
A



B



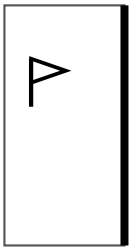
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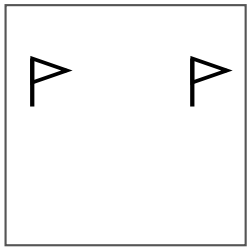


D

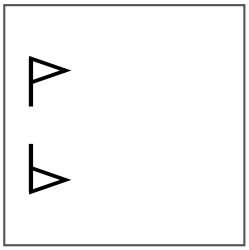
B. Fold, punch and unfold

Q4. A square paper is folded once (right half over the left half) and a hole is cut in the shape shown. The thick edge is the fold. Which figure shows the paper after unfolding?

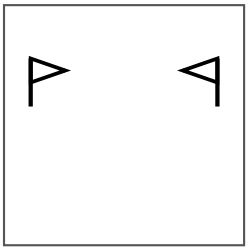




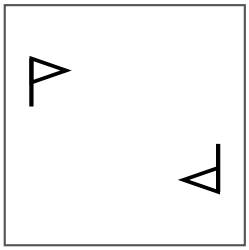
A



B



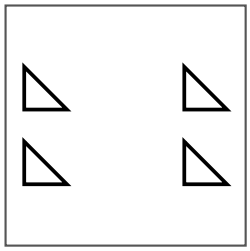
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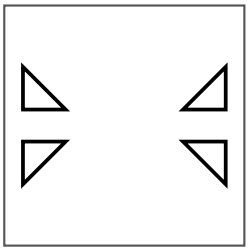
D

Q5. A square paper is folded twice (right over left, then bottom over top) and a hole is cut as shown in the folded quarter (thick edges are the folds). Which figure shows the paper after unfolding?

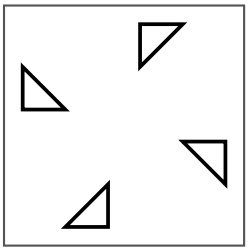




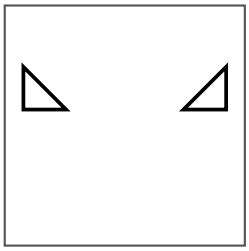
A



B



C



D

Q6. A square paper is folded twice (in half, then in half again) and one punch is made in the middle of the folded layers, away from every fold. How many holes appear when the paper is unfolded?

A. 4 B. 2 C. 3 D. 8

Answer key

Q	Answer	Reason
1	B	A fold is a mirror in the dotted line: the right-hand figures are flipped left-right and land in the left half (L at the same height, arrow turned from NE to NW). A leaves them out; C slides them across without flipping; D turns them through 180° (wrong, they would end upside-down).
2	D	A horizontal fold flips top and bottom: the top figures are reflected up-down into the bottom half (the flag keeps its direction but is turned upside-down, and the triangle's right angle moves to the top). A leaves them out; B slides them down unflipped; C turns them through 180°, which also reverses left and right.
3	A	Folding on the "/" diagonal swaps x and y: each lower-right figure is reflected into the upper-left part and the shape itself is reflected (the L and the triangle appear as mirror images). B turns the figures through 180° (a rotation, not a fold); C moves them without reflecting the shapes; D omits them.
4	C	Unfolding reflects the hole in the fold line: a left-right mirror copy appears on the right, at the same height. A has the copy in the right place but not flipped; B puts the copy below (wrong fold direction); D is a 180° copy.
5	B	Each unfolding is a reflection: the hole is reflected in the vertical fold and in the horizontal fold, so there are four mirror images (the copy opposite the original is turned 180°). A has the right positions but unflipped shapes; C has four copies related by rotation (a turn, not a fold); D has only two copies.
6	A	Two folds make four layers, and a punch through four layers makes 4 holes (2 × 2). With one fold there would be 2 holes; with three folds, 8.

CHAPTER 6

Quick Revision

One page for the last day

Fold line	Point goes to	Arrow
Vertical	(x, y) to (−x, y)	E and W swap
Horizontal	(x, y) to (x, −y)	N and S swap
Diagonal “/”	(x, y) to (y, x)	N and E, S and W, NW and SE swap
Diagonal “\”	(x, y) to (−y, −x)	N and W, S and E, NE and SW swap

Cut position	Unfolded result
Interior	Mirror copies (2^n after n folds)
On a fold edge	One whole symmetric shape (half-circle gives circle, triangle gives diamond)

PYQ RULE: All 4 PYQs: a fold is a reflection. Transparent sheet: reflect the folded part and overlap. Cut paper: unfold last fold first. Reverse: shapes on a middle line were on a fold edge.

ROUTINE: Exam-day routine: mark fold lines; reflect asymmetric parts; unfold in reverse order; test the symmetry; eliminate. 45 seconds per question. Paper Folding is the last topic named in the syllabus line.